
Data on Fire: A Hands-On Intro to Spark in Fabric

*SQL Saturday Oregon/SW
Washington 2025*



SQL Saturday 1130 Portland/Vancouver

November 15th, 2025



SQL Saturday 2025 Sponsors



Microsoft Fabric and Oregon Data (SQL Server) User Groups are teaming up in 2026!

Look for combined, in-person & hybrid meetings announced by both groups



[Register on Meetup](#) to join us



Get Fabric Certified by Dec 31!

Free & discounted exam vouchers &
training events
...but they are going fast!

DP-600 & DP-700 online study groups
hosted live by our local user groups



Information: community.fabric.microsoft.com

community.fabric.microsoft.com/t5/Power-BI-Community-Blog/Fabric-Data-Days-Get-Certified

FABCON
Microsoft Fabric
COMMUNITY CONFERENCE

SQLCON
Microsoft SQL
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Unveiling the **Future** of Data



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ATLANTA26
16-20 MARCH 2026

**Microsoft Fabric
and SQL Server
teaming up on the
big stage!**

[Register here](#)

Jason Romans

Cloud Data & Integration Developer

The DAX Shepherd



 @sql_jar

 jason-r-sql-jar

 <https://thedaxshepherd.com/>



 Nashville, TN, USA

 Began Career as a SQL Server DBA

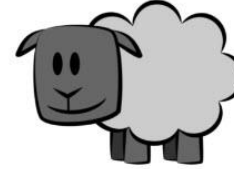
 Transitioned to Microsoft BI Stack

 Data Engineering to Data Modeling

 Infrequent Blogger

 Fan of Dimensional Models & Doctor Who

www.thedaxshepherd.com



The DAX Shepherd

Musings on the Microsoft BI Stack



[Home](#) [About Me](#) [Simple Talk](#) [Presentations](#) [A Speaker's Journey](#)

Presentations

Sessionize

This is my [Sessionize Profile](#) that has the conferences I have spoken at along with future events. It has a couple of my most popular sessions.

Presentation Slides

This is my [GitHub Repository](#) with the presentation slides for each event.

Recorded Sessions

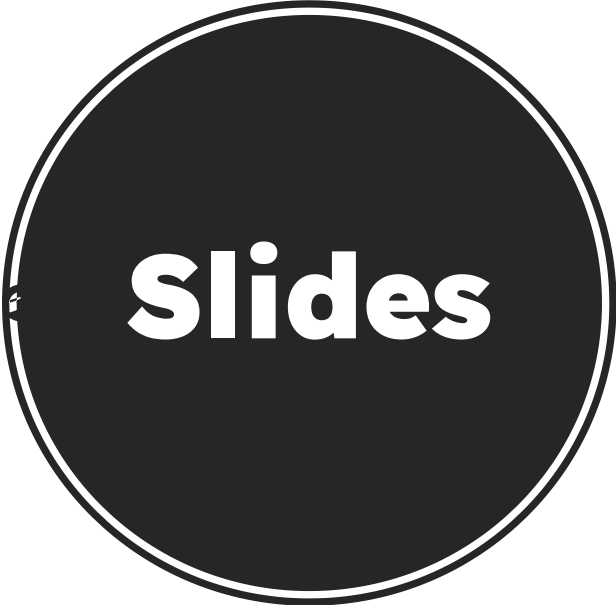
Simple Talks Podcast | Episode 4 – Coffee chat with Jason Romans

About Jason Romans



I love working with the Microsoft BI Stack. I am passionate about learning.

[A Speaker's Journey](#)



Slides

Shoulders of Giants



Our Journey



1. Intro

2. Python

3. PySpark

4. Uses

5. Conclusion

Our Journey



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What lit the fire for Apache Spark

The Netflix Prize

Began Oct 2006

Goal - improve Netflix's Cinematch
algorithm by at least 10%

Prize was 1 million dollars

Took until 2009

Couldn't the Elephant* Help?

* Hadoop's Mascot is an Elephant

Hadoop was not optimal for Machine Learning – multiple passes over disk

Need for new tooling

Shift to in-memory versus disk

Like Analysis Services Multi-Dimensional to Tabular

Contest must have led to work on Spark

Flashbacks of submitting homework digitally

Front runner BellKor's Pragmatic Chaos

Merger of teams from AT&A Labs and
Commendo Research

July 26, 2009 two teams met minimum
requirements

The Ensemble (Spark team) had a
better improvement in score

Lost by submitting 20 minutes later

What is Powered by Spark

- Apache Spark
- HDInsight
- Azure Synapse Analytics
- Databricks
- Microsoft Fabric

**Now let us
install
Spark**



Installing Spark

Step 1 of 42

```
$ pip install pyspark
No module named pip:
ModuleNotFoundError
Error importing setuptools module:
  Install' command is unavailable until setuptools is
  Ensure pip, setuptools, and wheel are up to date
  For "upgrade spark pip.ptgspec "ll ergk!
Upgrade pyspark --no-cache-dir setuptools wheel
Value for scheme.headers does not match
  to avoid this problem; \f errcode, with exame>
Retrying (Retry(total=4, connect=-4323 after Exceptio
  annotate; error: (versygtut), line 230, init _efor1
File '/usr/lib/python3.8/supprocs.py, line 231, in me
  trod self.s.connect(sockaddr cannect) timeout
File '/usr/lib/python3.8/socket.py', line 26, i meth
  self.s.connect(sockaddr)
Interna to an attack involve'next
Collecting pyspark _apack-3.2.1-bin-hadoopg.3.2.cg int
Downloading Apache-spark-3.2.1-i-nstaller
  error Value for scheme.headers does not match; ae' matc
  to avoridel tryh connect be found; > to avoid this problem
Retrying (Retry(total=<connect=,) after Exception annotate
-- again:z https://files.pythnonosted.org/poack,org)P/28n-
  (https:'files.pythonhosted.org(hjaps://packages/5565f.1
confirming) package failed: There was a p: 'crobiem-c t
Exception too problem confirming the ssl certificate: HTTPS
  (host='>>'https://p.jpu-1.jsom100ht> HTTPSConnectinPool(mo]
Could not install pagages due to an vo space on device
[Errno 28) nb insstall paccages due to an an an OSEserer:
```

Wait!
Microsoft
Fabric makes
this easy



Notebooks in Microsoft Fabric

Can apply to other environments

Notebook Gallery

https://community.fabric.microsoft.com/t5/Notebook-Gallery/bd-p/pbi_notebookgallery

There was a notebook contest (it is closed now):

<https://powerbi.microsoft.com/en-us/blog/introducing-the-first-ever-fabric-notebooks-competition-for-power-bi/>

Microsoft Fabric Architecture

- Data is stored in OneLake

ONELAKE

"THE ONEDRIVE
FOR YOUR DATA"

VISUALLY
EXPLAINED

EACH FABRIC ORGANIZATION HAS
ONE DATA LAKE, ENTIRELY MANAGED FOR YOU
IT'S UNIFIED ACROSS ALL REGIONS & YOU PAY
PER GB STORED (NO SCALING NEEDED).

ALL ANALYTICAL ENGINES CAN ACCESS ONELAKE DATA

SQL

SPARK

POWER BI

KQL DB

DATA LAKE + WAREHOUSE
= LAKEHOUSE

WITH FABRIC, YOUR
WAREHOUSE
DATA IS ALSO STORED
IN ONELAKE, AND
ACCESSIBLE IN AN OPEN
FORMAT

THIS IS A
LAKEHOUSE

IT ALLOWS YOU TO DISTRIBUTE
OWNERSHIP TO ORGANIZATIONS,
AND EASILY COLLABORATE ON
DATA

IT CONTAINS



A FILES
AREA



A TABLES
AREA

SHORTCUTS

ALLOW YOU TO QUERY
DATA FROM ONE LAKEHOUSE
TO ANOTHER WITHOUT DATA
MOVEMENT

SHORTCUTS LINK
TO OTHER CLOUDS
TOO!

THE FILES AREA CAN CONTAIN UNSTRUCTURED
& SEMI-STRUCTURED DATA.

IT'S NOT UNCOMMON TO ORGANIZE THEM IN
THREE "AREAS": BRONZE, SILVER & GOLD

ONELAKE

ADLS GEN2
SDK & APIs
CAN NATIVELY ACCESS ONELAKE

Microsoft Fabric Architecture

- Compute engines sitting on top of files
- Languages and compute
 - i.e. T-SQL with Warehouse

Compute & Language

 PySpark (Python) ▾

Spark

✓ PySpark (Python)

Spark (Scala)

Spark SQL

SparkR (R)

Python

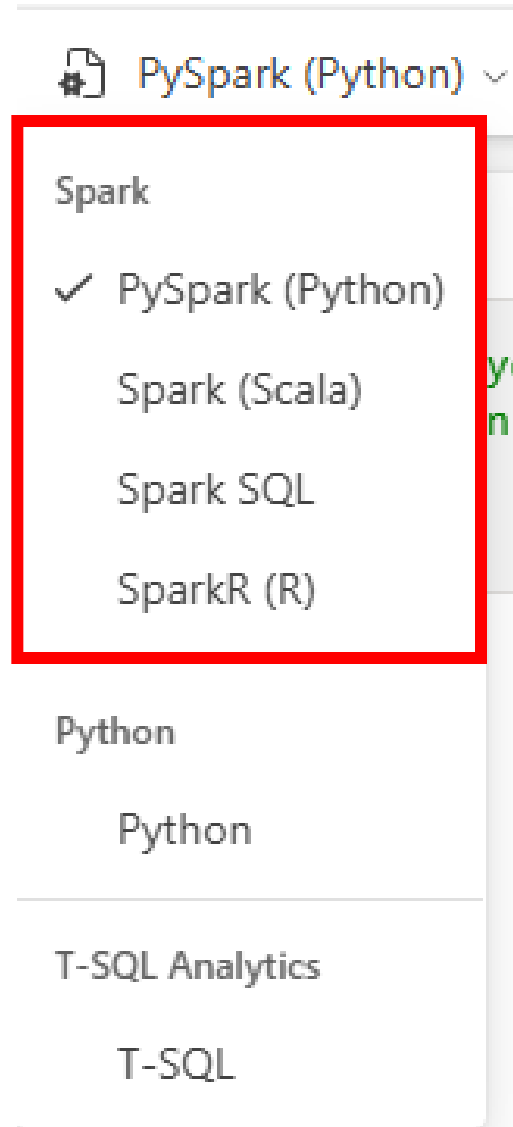
Python

T-SQL Analytics

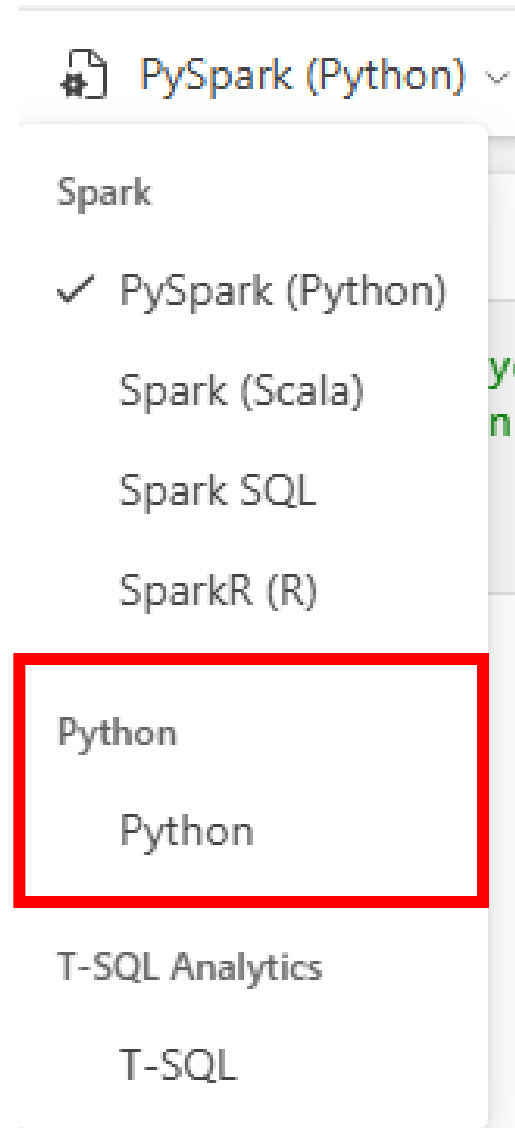
T-SQL

Spark

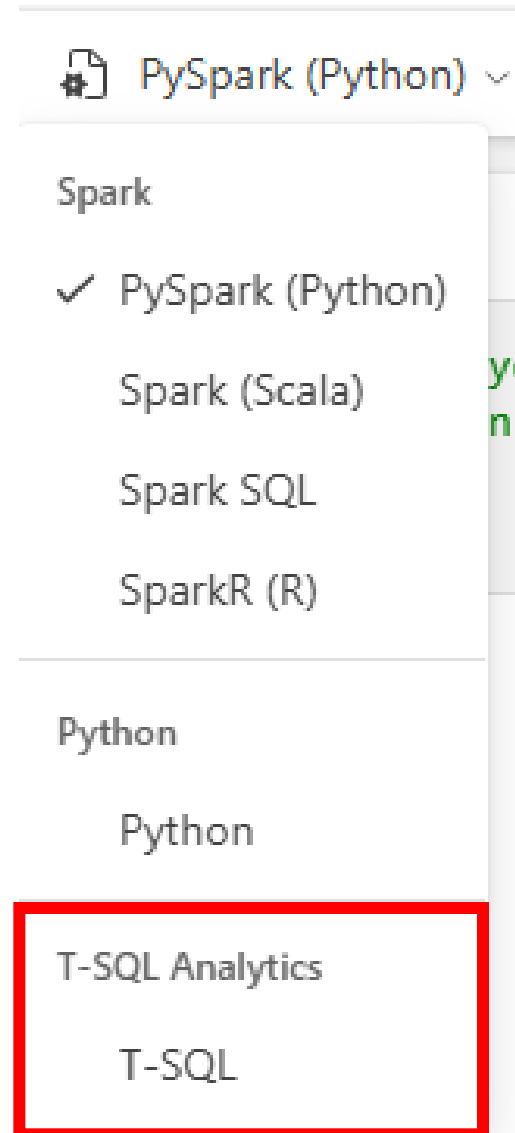
(Python, Scala, SQL, R)



Python (Python)



T-SQL Analytics (T-SQL)



Choosing PySpark or Python Compute (quick)

Scenario	Recommended Notebook
Includes pre-installed DuckDB and Polars libraries	Python Notebooks
Small to medium data (fits in memory)	Python Notebooks (or PySpark on single-node Spark cluster)
Rapid exploration & prototyping	Python Notebooks (or PySpark on single-node Spark cluster)
Large datasets (10GB+) exceeding memory	PySpark Notebooks
Complex data workflows or ETL pipelines	PySpark Notebooks
High-concurrency or parallel execution	PySpark Notebooks
Needs Spark-native APIs (MLlib, SQL, Streaming)	PySpark Notebooks

<https://learn.microsoft.com/en-us/fabric/data-engineering/fabric-notebook-selection-guide>

Choosing PySpark or Python Compute

Scenario	Python Notebooks (2-core VM)	PySpark Notebooks (Spark Compute)
Startup Time	The built-in starter pool initializes in approximately 5 seconds, while the on-demand pool takes around 3 minutes.	Start-up ranges from ~5 seconds (starter pool) to several minutes (on-demand Spark clusters).
Quick Transformations & API Calls	Ideal for small to medium sized datasets (up to 1GB)	Optimized for large datasets using vectorized execution.
Moderate Workloads	Not optimized for data sizes nearing memory saturation	Efficient at scaling via distributed compute.
Handling of Large Datasets	Limited by single-node memory. May struggle with scaling.	Distributed processing ensures scalable handling of multi-GB to TB workloads.
High-Concurrency Execution	Manual FIFO-style parallelism per notebook	System-managed concurrency with support for parallel execution.
Resource Customization & Scaling	Fixed compute (2-core VM); does not auto scale. Users can manually scale out using %%config within the notebook.	Flexible resource allocation; supports autoscaling and custom Spark configurations.

<https://learn.microsoft.com/en-us/fabric/data-engineering/fabric-notebook-selection-guide>

Type of Compute for Notebooks

- Spark Based
 - Cluster
- Single Node Python
 - 2 vCores, 16G RAM
- T-SQL Analytics
 - Warehouse

Python Notebook

- Has libraries installed for dealing with “small-big” data
 - Less than 10 Gigabytes
 - Fits in memory
- Example Libraries installed
 - Polars
 - DuckDB

Languages for Spark

Different choices of languages

Built with Scala

- PySpark (Python)
- Spark (Scala)
- Spark SQL
- SparkR (R)

 PySpark (Python) ▾

Spark

✓ PySpark (Python)

Spark (Scala)

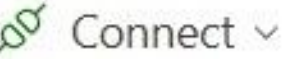
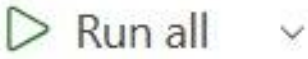
Spark SQL

SparkR (R)

Magic Commands – set language by cell

Magic command	Language	Description
%%pyspark	Python	Execute a Python query against Apache Spark Context.
%%spark	Scala	Execute a Scala query against Apache Spark Context.
%%sql	SparkSQL	Execute a SparkSQL query against Apache Spark Context.
%%html	Html	Execute a HTML query against Apache Spark Context.
%%sparkr	R	Execute a R query against Apache Spark Context.

HomeEditAI toolsRunViewCommentsHistoryDevelopShare



PySpark (Python) ...



»

Explorer

▼

Magic Commands

1%%spark

Spark (Scala) ▼

Our Journey



1. Intro

2. Python

3. PySpark

4. Uses

5. Conclusion

Python Language



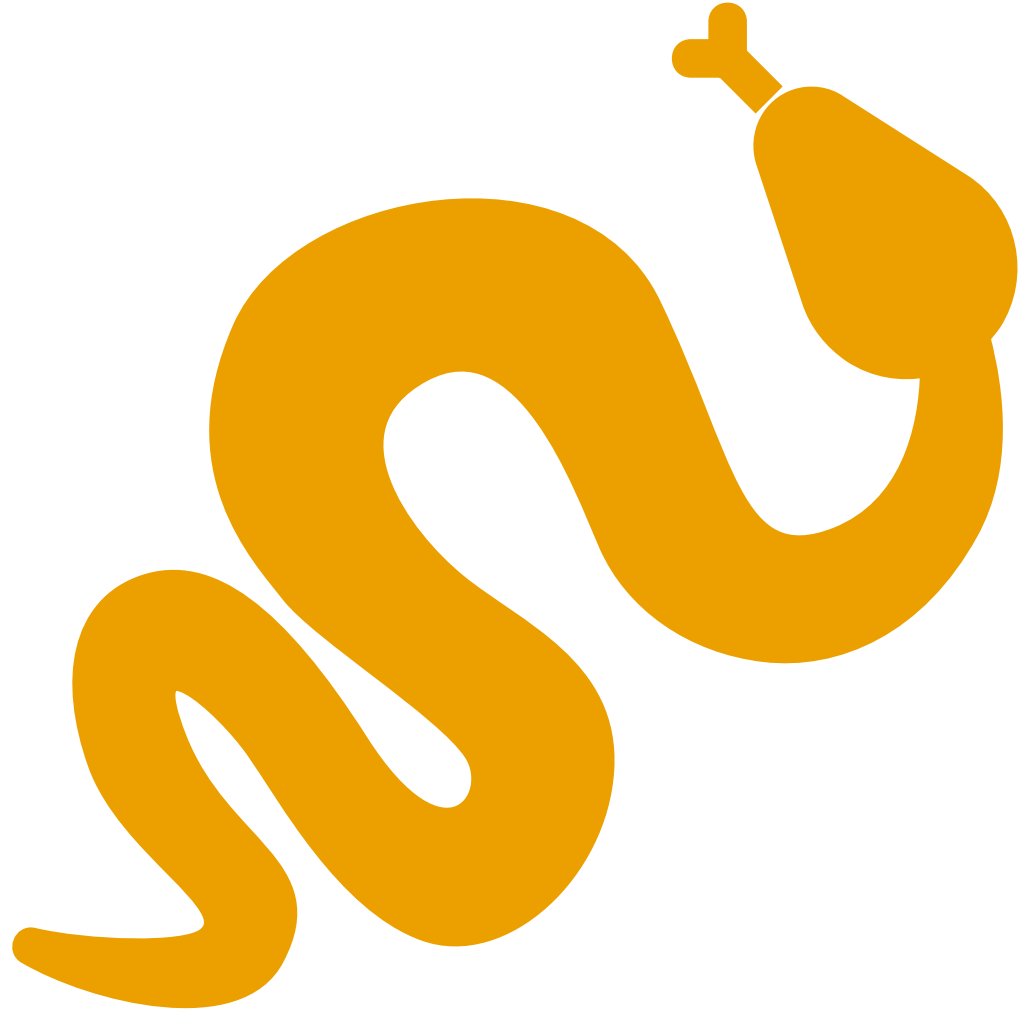
HOW YOU INTERACT
WITH SPARK



HOW YOU MANIPULATE
THE DATA

Python Demo

***You can thank me for not
having an actual picture of
a snake here*



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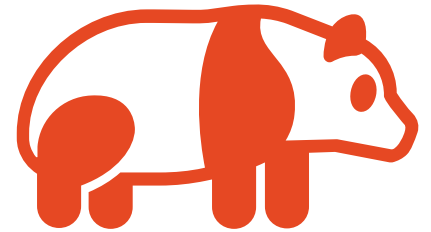
PySpark



Python API for
Spark



Most operations on
a DataFrame



Like Pandas but
distributed

DataFrame

- Resilient Distributed Dataset (RDD)
- Conceptually the same as a table
 - Abstraction
 - Rows
 - Columns

**Spark is
Lazy – in a
good way**



Lazy Evaluation

- Waits until an action is requested
- Actions
 - Counting number of rows in a Spark DataFrame
 - Showing output
 - Writing data to a file or data source
 - Transferring data from a Spark DataFrame to a native object in Python

Benefits of Lazy Evaluation

- Saves resources
- Plan can be optimized

Pandas (non-spark, historic) is eager evaluation

Fabric in Visual Studio Code



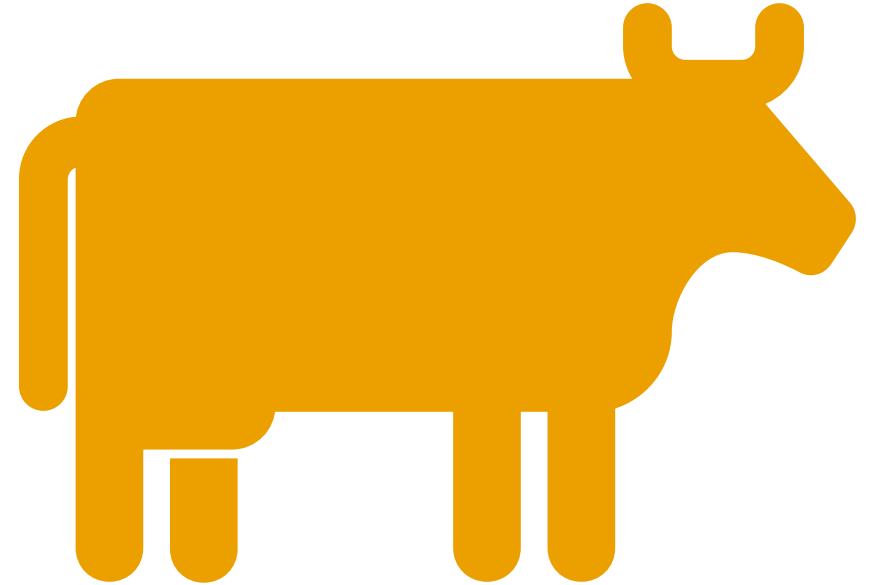
EDIT NOTEBOOKS



CONNECT TO COMPUTE
IN MICROSOFT FABRIC

Data Wrangler

**Like Power Query, but for
PySpark and Python**



PySpark Demo



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Uses



Big Data Processing - handling massive datasets



Machine Learning at Scale - training ML models on large data



Real-Time Stream Processing - processing live data streams



ETL & Data Pipelines - building data workflows



Advanced Analytics - complex calculations and BI



Data Exploration & Discovery - interactive data analysis

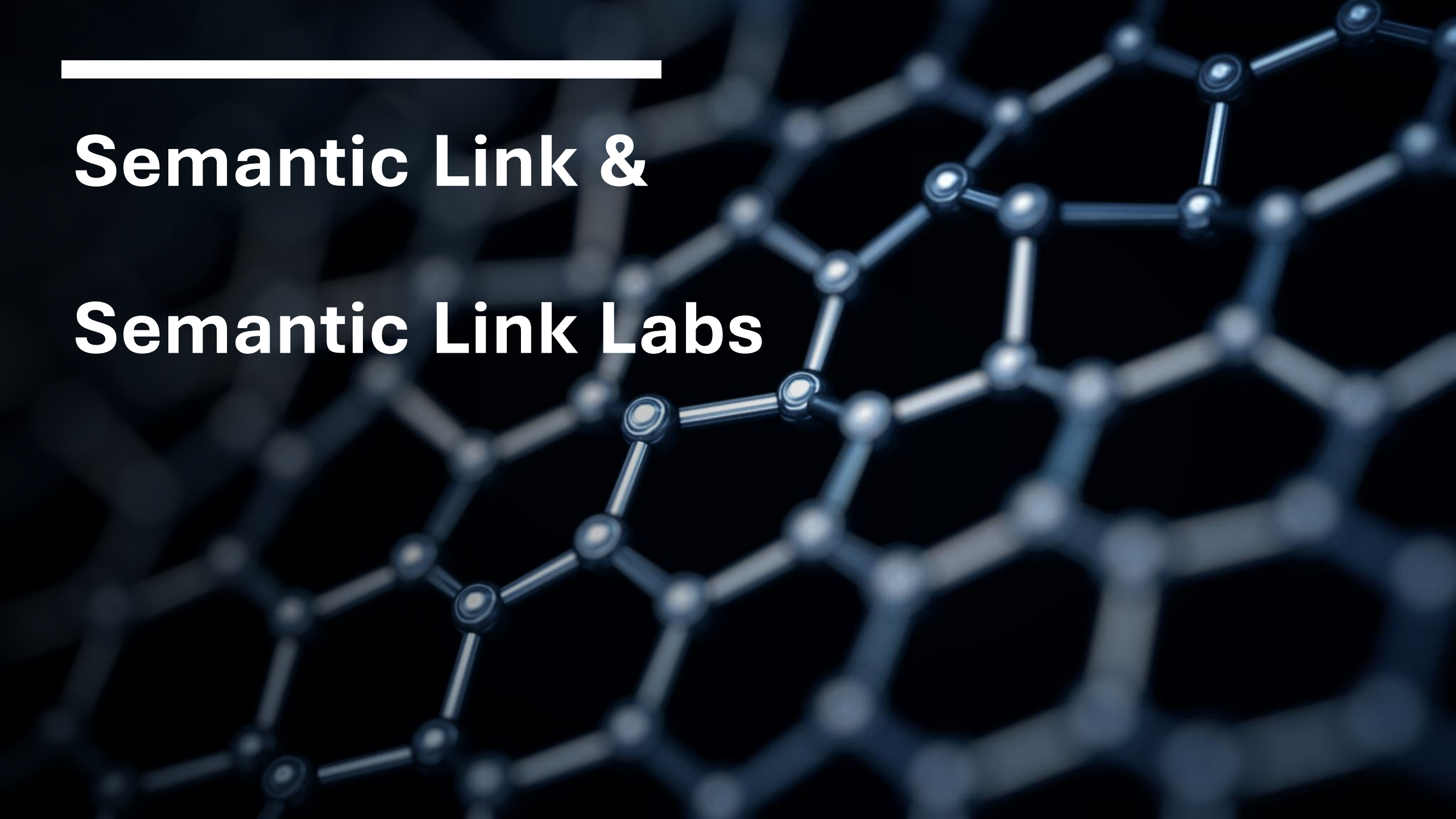
Executing Notebook using a Pipeline

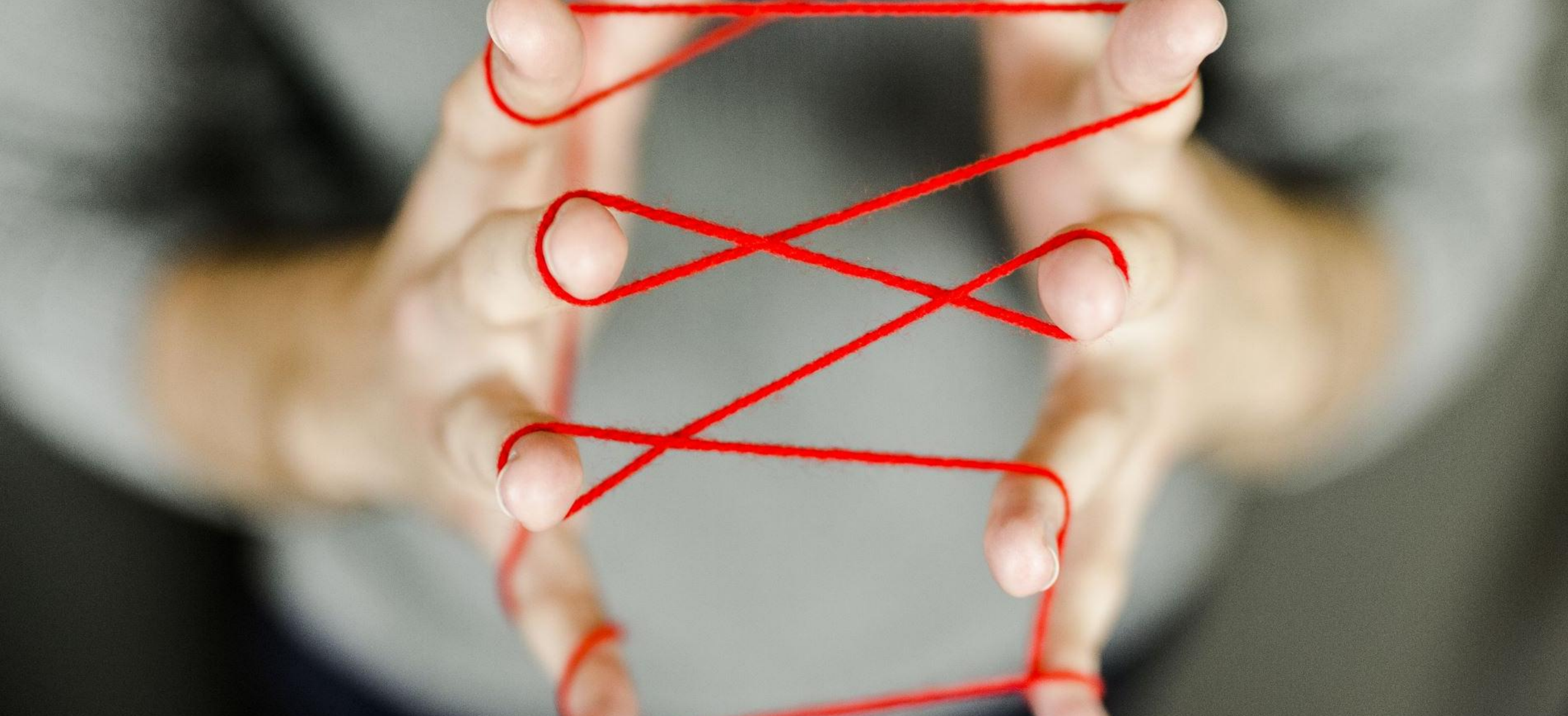
- Either
 1. Use Environment with Libraries
 - Microsoft recommended way
 - Only for Spark
 2. Python inline installation
 - Enable %pip install for pipeline, add "_inlineInstallationEnabled" as bool parameter equals True in the notebook activity parameters.

Reference:

- <https://learn.microsoft.com/en-us/fabric/data-engineering/library-management>

Semantic Link & Semantic Link Labs





Semantic Link Demo

Resources

Fabric Samples

- <https://github.com/microsoft/fabric-samples>
- Semantic Link
 - <https://learn.microsoft.com/en-us/fabric/data-science/semantic-link-overview>
- Semantic Link Labs
 - <https://github.com/microsoft/semantic-link-labs>

Resources

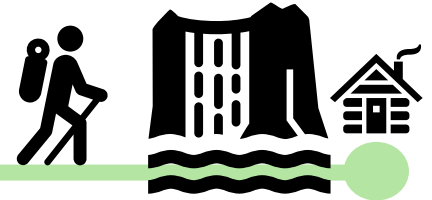
PySpark Book – Data Analysis with Python and PySpark

- <https://www.oreilly.com/library/view/data-analysis-with/9781617297205/>

PySpark Book – Intro to PySpark (Free HTML version)

- <https://pedropark99.github.io/Introd-pyspark/>

Our Journey



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Conclusion

Powerful and Scalable Platform

Apache Spark in Microsoft Fabric Notebooks provides a robust and scalable solution for handling big data analytics tasks efficiently.

User-Friendly Tools

The platform offers intuitive and practical tools that simplify data analysis for professionals of all skill levels.

Unlocking Valuable Insights

Understanding core concepts and leveraging these tools enables data professionals to extract meaningful insights effectively.

Complete Session & Event Surveys

SQL Saturday Oregon/SW
Washington Session Survey



SQL Server Usage Survey



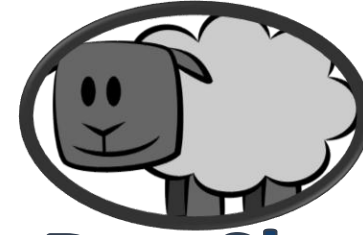
Session Evaluation



Data on Fire: A Hands-On Intro to Spark in Fabric

Thank you

Jason Romans
thedaxshepherd@gmail.com
www.thedaxshepherd.com



The Dax Shepherd

